

Tracking Targets through the Blind Doppler

Problem 2 — AE 6106: Bayesian Estimation for Tracking and Navigation

Tarshit Sehgal (22B0077)
Bhuvneshwar Rathore (24B2255)
Jai Aditya Garg (25M0017)
Tanjay Banikya (25M0022)

Radar measures three quantities:

$$z_k = \begin{bmatrix} x \\ y \\ \dot{r} \end{bmatrix}, \quad \dot{r} = \frac{x\dot{x} + y\dot{y}}{\sqrt{x^2 + y^2}}$$

Blind Doppler condition:

$$|\dot{r}| < L, \quad L = 300 \text{ km/h}$$

When range-rate falls below the threshold L , the **radar cannot distinguish the target from clutter**, no measurement is available.

Scenario

- Target starts at (0, 60 km)
- Initial speed: $v_0 = 800 \text{ km/h}$
- Turns induce blind Doppler intervals
- $T_k = 2 \text{ s}$, $N = 100$ steps

Challenge

Maintain track continuity *through* the blind interval and recover quickly after detection resumes.

Unscented Kalman Filter (UKF)

- No Jacobian required
- Propagates a minimal set of *sigma points* through the nonlinear $h(\cdot)$
- Captures nonlinear effects to 3rd-order accuracy
- Handles state-dependent R_k naturally
- Graceful degradation during blind intervals (prediction-only)

Particle Filter (PF)

- Fully nonlinear, non-Gaussian capable
- Represents posterior with weighted particles
- Two-population design handles blind Doppler explicitly
- ASIS mechanism seeds constrained particles at blind onset
- Each particle carries its own $R_i = R(\mathbf{x}^{(i)})$

Both filters go beyond linearization, a natural fit for the highly nonlinear range-rate measurement.

Member	Contribution
Tarshit Sehgal	Coding LKF, EKF, UKF and Presentation
Bhuvneshwar Rathode	Worked on LKF, EKF, UKF and PF and did filter performance analysis
Jai Aditya Garg	Filter performance analysis
Tanjoy Banikya	Worked on EKF and UKF

State vector: $\mathbf{x}_k = [x_k \quad \dot{x}_k \quad y_k \quad \dot{y}_k]^\top$

Coordinated Turn (CT) model: $\mathbf{x}_k = F_k(\omega) \mathbf{x}_{k-1}$

Flight phases:

$$F_k = \begin{bmatrix} 1 & \frac{\sin \omega T}{\omega} & 0 & -\frac{1 - \cos \omega T}{\omega} \\ 0 & \cos \omega T & 0 & -\sin \omega T \\ 0 & \frac{1 - \cos \omega T}{\omega} & 1 & \frac{\sin \omega T}{\omega} \\ 0 & \sin \omega T & 0 & \cos \omega T \end{bmatrix}$$

Process noise:

$$\mathbf{w}_k \sim \mathcal{N}(0, Q), \quad Q = \sigma_w^2 I_2, \quad \sigma_w = g/\sqrt{T_k}$$

Phase	Steps	ω
1	1–45	0
2	46–51	$-3g/v_0$
3	52–64	0
4	65–70	$+3g/v_0$
5	71–100	0

Phase 2: right turn

Phase 4: left turn

Nonlinear measurement function:

$$h(\mathbf{x}) = \begin{bmatrix} x \\ y \\ \frac{x\dot{x} + y\dot{y}}{\sqrt{x^2 + y^2}} \end{bmatrix}$$

State-dependent measurement noise:

$$R_k = R(\mathbf{x}_k), \quad (\text{function of } r, \theta)$$

$$\lambda = e^{-\sigma_\theta^2/2}$$

Filter initial conditions:

Parameters:

Parameter	Value	Description
T_k	2 s	Sampling interval
σ_r	250 m	Range std. dev.
σ_θ	1°	Angle std. dev.
$\sigma_{\dot{r}}$	3 m/s	Range-rate std. dev.
σ_w	$g/\sqrt{T_k}$	Process noise
L	300 km/h	Blind threshold
v_0	800 km/h	Initial speed

Linearized Kalman Filter (LKF)

Key idea: Linearize $h(\mathbf{x})$ *once* at a fixed nominal point; reuse H every step.

Fixed Jacobian (evaluated at $\mathbf{x}_0 = [0, 0, 60000, -v_0]^\top$):

$$H = \left. \frac{\partial h}{\partial \mathbf{x}} \right|_{\mathbf{x}_0}$$

CV prediction model (filter assumes Constant Velocity):

$$\hat{\mathbf{x}}_k^- = F\hat{\mathbf{x}}_{k-1}, \quad P_k^- = FP_{k-1}F^\top + GQG^\top$$

Update:

$$K_k = P_k^- H^\top (HP_k^- H^\top + R_k)^{-1}$$
$$\hat{\mathbf{x}}_k = \hat{\mathbf{x}}_k^- + K_k(z_k - H\hat{\mathbf{x}}_k^-), \quad P_k = P_k^- - K_k(HP_k^- H^\top + R_k)K_k^\top$$

Limitation

H is **never re-evaluated** — linearization error accumulates as the target maneuvers. Performs poorly during turns.

Extended Kalman Filter (EKF)

Key idea: Re-linearize $h(\mathbf{x})$ at the *current predicted state* $\hat{\mathbf{x}}_k^-$ at every step.

Prediction (same CV model as LKF):

$$\hat{\mathbf{x}}_k^- = F\hat{\mathbf{x}}_{k-1}, \quad P_k^- = FP_{k-1}F^\top + GQG^\top$$

Update with fresh Jacobian:

$$H_k = \left. \frac{\partial h}{\partial \mathbf{x}} \right|_{\hat{\mathbf{x}}_k^-}, \quad \hat{z}_k^- = H_k \hat{\mathbf{x}}_k^-$$

$$K_k = P_k^- H_k^\top \left(H_k P_k^- H_k^\top + R_k \right)^{-1}$$

$$\hat{\mathbf{x}}_k = \hat{\mathbf{x}}_k^- + K_k (z_k - \hat{z}_k^-), \quad P_k = P_k^- - K_k \left(H_k P_k^- H_k^\top + R_k \right) K_k^\top$$

Improvement over LKF

Fresh H_k at every step reduces linearization error during maneuvers. R_k evaluated at $\hat{\mathbf{x}}_k^-$ (state-dependent noise handled consistently).

Parameters: $n_x = 4$, $\kappa = 1$, $\rho = 5$

$$\nu_0 = \frac{\kappa}{\rho} = 0.2, \quad \nu_i = \frac{1}{2\rho} = 0.1$$

Sigma Points ($2n_x + 1 = 9$ total):

$$\chi_0 = \hat{\mathbf{x}}, \quad \chi_i = \hat{\mathbf{x}} \pm \sqrt{\rho} \ell_i$$

ℓ_i : i -th col. of $\text{chol}(P)$. Regularisation:

$$P \leftarrow \frac{P+P^\top}{2} + 10^{-9}I.$$

Propagate & Predict:

$$\chi_i^* = F\chi_i, \quad \hat{\mathbf{x}}_k^- = \sum_i \nu_i \chi_i^*$$

$$P_k^- = \sum_i \nu_i (\chi_i^* - \hat{\mathbf{x}}_k^-)(\cdots)^\top + GQG^\top$$

CV Model Mismatch

Filter predicts with F (Constant Velocity). True target follows CT model. Process noise $GQG^\top$ compensates only partially.

Blind Doppler

If $|\dot{r}| < L$: skip update.

Set $\hat{\mathbf{x}}_k = \hat{\mathbf{x}}_k^-$, $P_k = P_k^-$.

Measurement sigma points:

$$\gamma_i = h(\chi_i^*), \quad \hat{z}_k = \sum_i \nu_i \gamma_i$$

Innovation covariances:

$$P_{zz} = \sum_i \nu_i (\gamma_i - \hat{z}_k)(\cdots)^\top + R_k$$

$$P_{xz} = \sum_i \nu_i (\chi_i^* - \hat{\mathbf{x}}_k^-)(\gamma_i - \hat{z}_k)^\top$$

Update:

$$K = P_{xz} P_{zz}^{-1}$$

$$\hat{\mathbf{x}}_k = \hat{\mathbf{x}}_k^- + K(z_k - \hat{z}_k)$$

$$P_k = P_k^- - K P_{zz} K^\top$$

Advantage

No Jacobian required. Captures nonlinear $h(\cdot)$ to 3rd-order accuracy via sigma points.

Modifications

- R_k evaluated at $\hat{\mathbf{x}}_k^-$
- $P_0 = \text{diag}(1000^2, 10^2, 1000^2, 10^2)$
- Cholesky stabilisation at each step

Motivation: Standard PF collapses during Blind Doppler — likelihood undefined when $|\dot{r}| < L$.

Pop. U ($N_U = 2000$, unrestricted):

$$\mathbf{x}_k^{(i)} = F\mathbf{x}_{k-1}^{(i)} + G\mathbf{w}^{(i)}$$

Pop. BZ ($N_{BZ} = 2000$, $|\dot{r}| \leq L$): via rejection sampling.

ASIS — spawn BZ at blind onset:

$$\pi_i = \begin{cases} 1 & |\dot{r}(\mathbf{x}^{(i)})| \leq L \\ 0 & \text{otherwise} \end{cases}$$

Normalise $\pi_i \rightarrow$ resample seeds from $U \rightarrow$ rejection-sample valid $\mathbf{x}_i^{BZ}$.

Mode switching: $p_U + p_{BZ} = 1$

$$\begin{cases} p_U = 1, p_{BZ} = 0 & \text{detected} \\ p_U = 0, p_{BZ} = 1 & \text{blind} \end{cases}$$

$$\hat{\mathbf{x}}_k^{PF} = p_U \bar{\mathbf{x}}_U + p_{BZ} \bar{\mathbf{x}}_{BZ}$$

Modifications

Hard switching (no fractional mixing).
No iteration cap on rejection sampling.

Likelihood update (detection phase):

$$w^{(i)} \propto \exp\left(-\frac{1}{2}(z_k - h(\mathbf{x}^{(i)}))^{\top} R_i^{-1}(z_k - h(\mathbf{x}^{(i)}))\right)$$

Each particle uses its own $R_i = R(\mathbf{x}^{(i)})$.

Stratified resampling:

$$c_j = \sum_{i=1}^j w^{(i)}, \quad u \sim \mathcal{U}(0, 1) \Rightarrow w^{(i)} \leftarrow \frac{1}{N_U}$$

Output & covariance:

$$\hat{\mathbf{x}}_k^{PF} = \sum_i w^{(i)} \mathbf{x}^{(i)}$$

$$P_k^{PF} = p_U P_U + p_{BZ} P_{BZ}$$

Key Feature

Per-particle R_i accounts for state-dependent measurement noise — more accurate than a single global R .

Limitations

- Particle degeneracy after resampling
- High cost: $\mathcal{O}(N)$ per step
- Risk of infinite loop in rejection sampling

LKF — fixed linearization:

$$\mathbf{x}_0^* = \begin{bmatrix} 0 \\ 0 \\ 60000 \\ -v_0 \end{bmatrix} = \begin{bmatrix} 0 \text{ m} \\ 0 \text{ m/s} \\ 60 \text{ km} \\ -222.2 \text{ m/s} \end{bmatrix}$$

The Jacobian H is computed *symbolically* using `jacobian()` in MATLAB and then evaluated numerically at $\mathbf{x}_0^*$.

EKF — rolling linearization:

$$\mathbf{x}_k^* = \hat{\mathbf{x}}_k^- \quad (\text{predicted state at each step})$$

H_k recomputed numerically at every time step via `H_func(x_est)`.

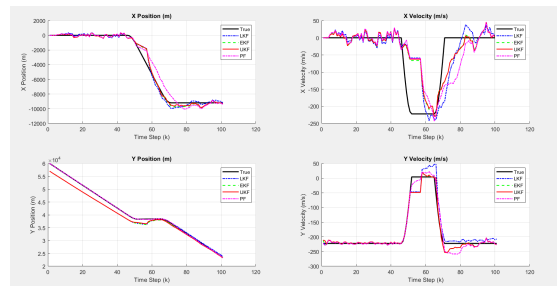
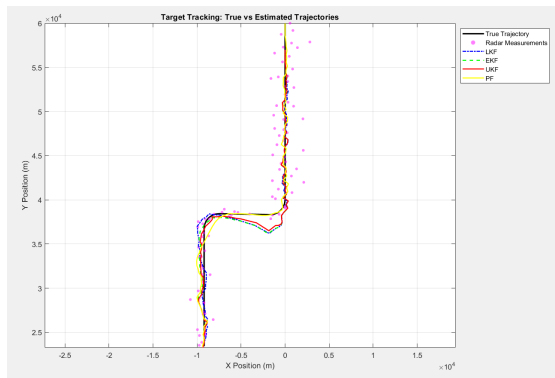
Why this nominal point?

- True initial state is $(0, 0, 60 \text{ km}, -v_0)$
- Target moves mostly in the y -direction initially
- $\dot{r} \approx -v_0$ at $t = 0$ (far above threshold L)
- Jacobian is well-conditioned at this point

Note

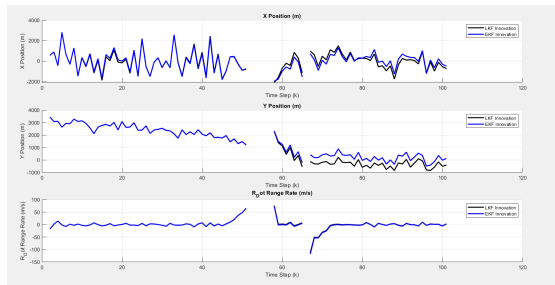
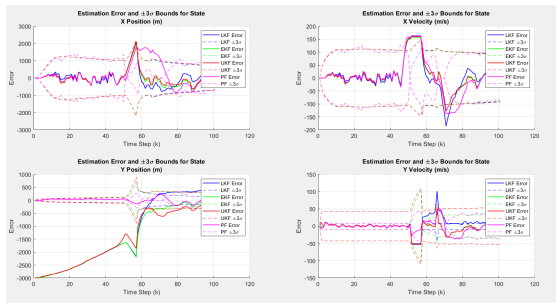
LKF linearization becomes increasingly inaccurate as the target turns away from $\mathbf{x}_0^*$.

Results — Estimated Trajectories & State Estimates



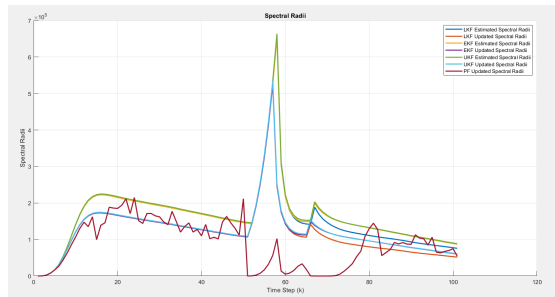
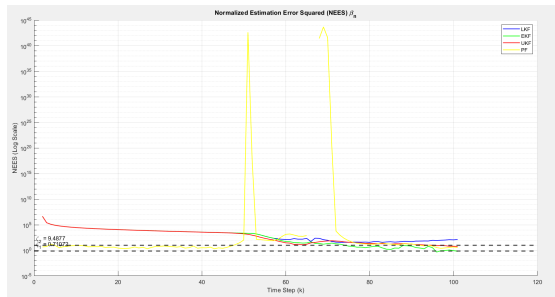
- All four filters track the true trajectory closely; LKF, EKF, UKF, PF nearly overlap
- All filters undershoot the turn slightly; CV model cannot capture CT dynamics
- LKF overshoots X velocity during the turn ($k \approx 46-64$); PF and UKF recover faster post-turn

Results — Error Bounds & Innovations



- Large error spike at $k \approx 55-60$ (first blind interval) across all states and filters
- UKF Y-position error grows steadily before the blind interval; CV/CT mismatch accumulates
- Innovations spike at blind interval boundaries; LKF Y-innovation has a persistent non-zero bias post-turn

Results — NEES & Spectral Radii



- **PF NEES spikes twice** ($k \approx 52$ and $k \approx 67$, up to 10^{45}); particle collapse at both blind onsets
- UKF NEES stays above ζ_2 for most of the run (95% violations) due to poor initial covariance
- Spectral radii peak sharply at $k \approx 57$; PF updated radii (dark red) are erratic throughout

Results — RMSE & NEES Comparison

State	LKF	EKF	UKF	PF
X Position (m)	466.09	429.67	426.42	694.57
X Velocity (m/s)	62.26	55.63	56.79	67.16
Y Position (m)	1828.4	1825.1	1796.4	87.34
Y Velocity (m/s)	21.10	16.31	16.35	15.38
NEES violations	100%	75%	95%	47%

Observations

- **EKF** lowest X velocity RMSE; **UKF** lowest X position RMSE
- **PF** dramatically best in Y position (87 vs 1800+ m); particle diversity survives the blind interval
- **PF** also best in Y velocity and NEES

Consistency (NEES)

- All filters exceed the 5% threshold
- **UKF**: 95% violations; large initial P_0 causes persistently high NEES
- **LKF**: 100%; fully inconsistent throughout
- **PF**: best at 47% despite collapse spikes

Key Findings

- 1 **Blind Doppler** forces predict-only; all filters degrade; quality depends on process model.
- 2 **LKF**: 100% NEES violations; fixed linearization fails entirely during turns.
- 3 **EKF**: best X -velocity RMSE; rolling re-linearization helps but CV mismatch remains.
- 4 **UKF**: best X -position RMSE but 95% NEES violations due to covariance inflation.
- 5 **PF**: best Y -state RMSE and NEES (47%); particle diversity handles non-Gaussianity but collapses at both blind onsets.

Limitations

- CV prediction vs. true CT motion; systematic mismatch inflates all errors
- PF collapses at *both* blind intervals; no iteration cap on rejection sampler
- UKF initial P_0 choice inflates NEES throughout

Future Work

- IMM filter for CV/CT mode switching
- Adaptive process noise Q_k
- Iteration cap + fallback for PF rejection sampler